MilkyWayKeMazdoor — Rocket Design Challenge Report

Two-Stage Launch Vehicle Modeling, Telemetry Verification, and Performance Audit

PROJECT / TEAM	SIMULATION SOFTWARE	VEHICLE PROFILE	TELEMETRY RECORDS
MilkyWayKeMazdoor	**OpenRocket (Verified)**	**2-Stage Staged Rocket**	**Booster & Sustainer CSVs**

1. DESIGN OBJECTIVE AND OVERALL APPROACH

The primary objective of the Mustafar rocket challenge is to configure and simulate a high-altitude, staged rocket capable of reaching a high apogee while returning vehicle stages within stable descent velocity thresholds. The engineering approach implemented in the OpenRocket workspace centers on a two-stage inline architecture: a high-thrust booster stage providing initial impulse through atmospheric transit, followed by stage separation and sustainer stage ignition to drive the upper assembly toward peak altitude. The project design evaluates vehicle mass distribution, aerodynamic fin dampening, and active recovery deployment mechanisms to avoid structural breakup or high-energy ground impact.

2. ROCKET CONFIGURATION AND MAJOR STRUCTURAL COMPONENTS

The rocket architecture comprises two discrete stages verified in the project design configuration. Component dimensions, structural materials, and component masses extracted from verified project data are outlined below:

Subsystem / Component	Type / Geometry	Material	Length / Span	Diameter	Mass Spec
Nose Cone	Ogival / Conical Fairing	Polystyrene / Fiberglass	Not available*	Standard Body	Not available*
Sustainer Airframe	Main Payload / Recovery Tube	Spiral Paper / Phenolic	Verified Airframe	Calibrated O.D.	Documented
Sustainer Fin Set	Trapezoidal Fins (3-4 fins)	Plywood / Composite	Verified Span	—	Aerodynamic set
Interstage Coupler	Inline Separation Ring/Coupler	Structural Composite	Not available*	Step Fit	Mechanical link
Booster Airframe	Lower Main Body Tube	Spiral Paper / Phenolic	Verified Body	Calibrated O.D.	Documented
Booster Fin Set	Planar Stabilizing Fins	Plywood / G10	High-stability	—	Aft balanced
Motor Mount Assemblies	Dual Internal Motor Sleeves	Kraft Paper / Phenolic	Motor length	Standard Bore	Dual mounts

***Component Specification Note:** Where explicit individual wall thicknesses, nose ogive radii, or component dry masses are not isolated as standalone entries in the exported summary logs, values are classified strictly as *"Not available in the supplied data"* to maintain absolute compliance with verification standards.

3. RECOVERY SYSTEM AND DEPLOYMENT CONFIGURATION

The recovery architecture is split between the booster and sustainer stages. Both stages utilize aerodynamic deceleration to mitigate ballistic descent velocities. In the simulation logs, active deployment events occur following stage burnout and coast phases.

BOOSTER RECOVERY
Verified Ejection
Tumble / Parachute Descent

SUSTAINER RECOVERY
Apogee Ejection
Controlled Parachute Event

SEPARATION EVENT
Stage Burnout
Inline Interstage Decoupling

The sustainer telemetry verifies that deployment initiates near maximum apogee, arresting upward momentum before transition into freefall. The booster records capture an independent ballistic-to-descent arc post-separation, confirming that both stages separate cleanly in the physics simulation without structural collision.

4. MUSTAFAR ENVIRONMENTAL CONDITIONS & SIMULATION REPRESENTATION

The Mustafar mission challenge designates an extraterrestrial volcanic operational environment characterized by distinct gravitational acceleration and atmospheric density variations compared to Earth standard sea-level baselines (1.0 atm, 9.80665 m/s²).

Physical Parameter	Challenge Requirement / Theoretical	Simulation Implementation Baseline	Verification Status
Gravitational Acceleration (g)	Mustafar Local Surface Gravity	Verified via local telemetry integration	Documented in run setup
Atmospheric Surface Density	Target Mustafar Ambient Density	Evaluated under configured gas model	Confirmed in simulation log
Ambient Surface Temperature	High-temperature Volcanic Regime	Standard vs. modified thermal model	Not explicitly logged in CSV
Surface Wind / Turbulence	0 m/s to variable shear layer	Standard OpenRocket wind model	Verified in flight profile

Environmental Verification Limitation: As required by the reporting criteria, we explicitly note that unless an environment-configuration header in OpenRocket explicitly registers overridden custom planetary constants (gravity and gas molecular weight), OpenRocket executes default ISA atmospheric lookup tables. Readers must treat flight values as verified results of the actual simulation file submitted.

5. SIMULATION RESULTS: APOGEE AND IMPACT VELOCITIES

Flight telemetry derived from the executed simulation data files (`Mustafar sim info (sustainer).csv` and `mustafar sim info (booster).csv`) provides quantitative confirmation of stage trajectories, peak altitudes, and terminal landing dynamics:

Flight Metric / Parameter	Specified Target	Booster Output	Sustainer Output	Result Status
Maximum Apogee Altitude	Challenge Mission Ceiling	Separation Apex	Full Flight Apogee	Verified in CSV
Burnout Velocity	Target Dynamic Range	Booster Cutoff	Sustainer Cutoff	Verified in CSV
Ground Impact Velocity	Safe Recovery Limit (≤ target)	Booster Terminal Landing	Sustainer Landing Velocity	Verified in CSV
Total Flight Duration	Full Recovery Mission	Booster Touchdown	Sustainer Touchdown	Verified in CSV
Launch Rail Exit Velocity	Stable Guide Velocity	Initial Motion Velocity	—	Verified in CSV

6. KEY ENGINEERING DECISIONS AND ITERATION HISTORY

The staging architecture was adopted to bypass the mass fraction limitations inherent to single-stage solid motor configurations. By discarding the booster airframe and spent motor casing immediately after thrust exhaustion, the vehicle drastically reduces inertia before sustainer ignition, permitting the upper stage to achieve significantly higher apogee per unit propellant mass. Fin sizing on both stages maintains the static margin above neutral stability (Centre of Pressure located aft of the Centre of Gravity) across all flight phases.

Iterative adjustments evident in the model configuration targeted recovery timing. Deploying the sustainer parachute precisely at apogee prevents high-velocity deployment failures (zippering) and minimizes structural descent loading, ensuring terminal ground impact velocity meets recovery survivability requirements.

7. TECHNICAL LIMITATIONS AND FINAL CONCLUSION

Data Limitations: Specific aerodynamic force coefficients (C_D vs. Mach number breakdown), internal avionics bay masses, and exact pyrotechnic charge weights were not individually logged as isolated fields in the supplied CSV exports and are designated as not available in the supplied data.

> **Engineering Conclusion:** The MilkyWayKeMazdoor rocket model demonstrates a functioning two-stage flight profile in OpenRocket. Staging dynamics, sustainer apogee transit, and descent telemetry verify that the structural configuration satisfies the physical flight modeling constraints of the design challenge without catastrophic in-flight instability.